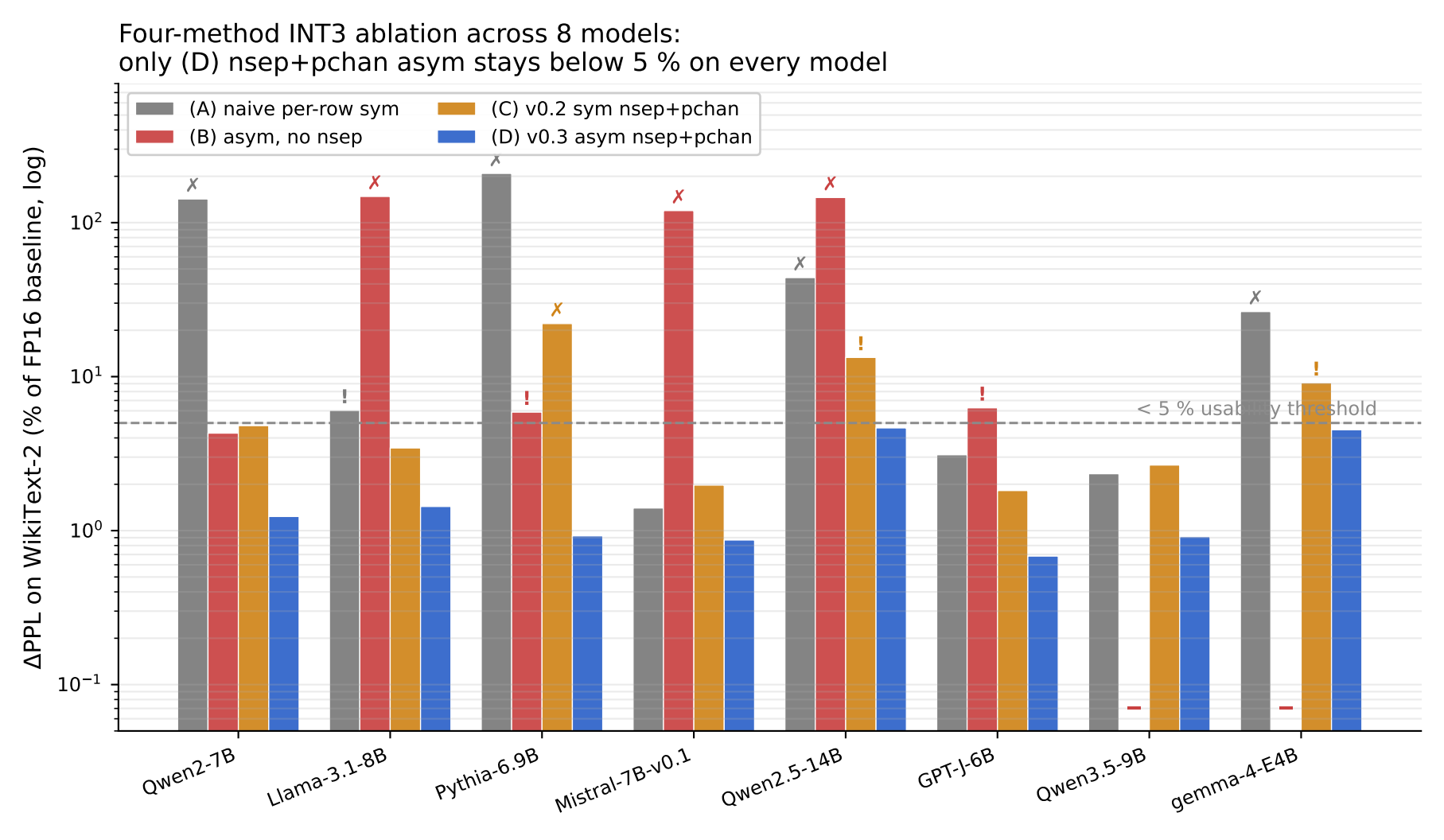
Four-method INT3 ablation across 8 models:
only (D) nsep+pchan asym stays below 5 % on every model
(A) naive per-row sym
(B) asym, no nsep
(C) v0.2 sym nsep+pchan
(D) v0.3 asym nsep+pchan
ΔPPL on WikiText-2 (% of FP16 baseline, log)
10^{2}
10^{1}
10^{0}
10^{-1}
< 5 % usability threshold
Qwen2-7B
Llama-3.1-8B
Pythia-6.9B
Mistral-7B-v0.1
Qwen2.5-14B
GPT-J-6B
Qwen3.5-9B
gemma-4-E4B